THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

May 17, 2018 14:30 pm – 17:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. The university is planning to convert an existing residential building into a hall of residence for students. In the absence of any experience in major renovation work, they are considering appointing a management contractor to represent their interests. As the engineering consultant of this project, explain to the university:

   a) the concept of management contracting with the aid of suitable diagram(s); and [13 marks]

   b) whether traditional procurement method in which consultant engineer is responsible to lead the project would be more appropriate when applied to this projects. [12 marks]

2. a) Explain with examples what are 'patent error', 'front / rear loading' and 'qualification' in tendering sense and how should the consultant engineer deal with these tendering issues? [10 marks]

   b) As shown in the tender breakdown below, the lump sum tender price of the lowest tenderer is erroneous. While the client intends to award the contract to this tenderer at the correct sum of HK$62,318,031, advise the client what is the most suitable practice and procedure to deal with this situation if tenderers are not allowed to correct genuine errors. [15 marks]

	HK$
Preliminaries	7,659,570
Substructure	5,675,210
Superstructure	45,098,343
External Works	934,908
Prime Cost and Provisional Sums	2,950,000
Total carried to form of tender:	65,318,031

3. a) An investor is planning to build a 300-room service apartment. The usable floor area of the service apartment will be 9,000m^2 and the circulation area will take up 20% of the total floor space. The construction cost is estimated to be $10,000/m^2 and it will take 24 months to construct. It is expected that the service apartment will have the vacancy rates of 40%, 20% and 0% during the first, second and third year of operation respectively at a rate of $1,000/apartment/day. The initial yield for similar service apartment investments would be 8%. With the cost of advertisement being budgeted at $1,000,000 and a short-term finance at 12% per annum, calculate the maximum site value for this development. [18 marks]

   b) Describe the major drawbacks of residual method and under what circumstances should discounted cash flow method be used for investment appraisal. [7 marks]

4. a) While planning for the construction of a 10-storey building (ground floor being included), it is found that the layout and size of each unit including those at ground floor are identical. Under this circumstance, what are the key considerations in terms of programming and how should the identified issues be addressed? [7 marks]

b) According to the contract, the project has to be completed within 270 days. With two units per floor, the proposed work sequence and time for construction of one residential unit are as follows:

Activity	*Duration (week)*
A	1
B	1
C	2
D	1
E	1
F	1
G	1
H	1
I	1

Based on the above data, calculate how long should be the 'turnover' for the activities. [5 marks]

c) Prepare a programme (in week) based on the data and findings in sub-sections (a) and (b). [8 marks]

d) Would the 'turnover' identified in sub-section (b) lead to an optimum productivity? If not, how can efficiency be improved? [5 marks]

5. a) The contractor has submitted a cost estimate for an additional work as specified in the variation orders. The cost estimate related to the new 150mm thick concrete slab is shown in Appendix I. As the Project Engineer, check the cost estimate and prepare a revised estimate based on your professional knowledge and judgement. A preliminary enquiry reveals that the price of materials and labour wages as shown in the contractor's cost estimate are in line with those in the market. [20 marks]

b) Define the term markup in the context of cost estimation and describe how markup is reflected in a tender. [5 marks]

Insitu Concrete (1:2:4, 40mm Aggregate), 150mm Thick Slab

Description			$/m^3	$
Material				
Cement @ $800.00/tonne			800.00	
Sand @ $120.00/tonne			120.00	
Aggregate $100.00/tonne			100.00	
			1,020.00	
Add	10% Inflation (due to possible price increase)		102.00	
			1,122.00	1,122.00
Plant			$/m^3	
Mixer, hired (ouput of mixer = 3 m^3/hour) @ $150/hour			150.00	
Crane and skip - $3,000/hour			3,000.00	
		Total plant cost =	3,150.00	3,150.00
Labour			$/m^3	
Mixer driver				
1 number x $100.00/hour			100.00	
Gang		$/hr		
2 x placers @ $100.00/hr		200.00		
2 x levellers @ $100.00/hr		200.00		
1 x vibration @ $120.00/hr		120.00		
		520.00		
Gang output = 1.17 hours / m3				
$520.00/hour ÷ 1.17 hours			444.44	
		Total labour cost =	544.44	544.44
		MATERIAL + PLANT + LABOUR :		4,816.44
		150mm thick slab :		722.47
		Unit Rate:		**$722.47/m^2**

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